DAY – 2

Topics – Free toolchain setup : Python , VS code , Jupyter /Colab free tier , OpenCV , Numpy , Matplotlib and Git

Python:

- To use Python on your system install a specific version or newest version of python .
- Consists of multiple modules , packages , these modules can be user defined or predefined.
- Using import statements for importing multiple packages
- The user defined modules needs to be installed before using pip command
- On pypi all the packages are present , all the package are present on pypi , search for the specific package
- To install any specific package , the command to be used is:

```
(venv) rjekker@psdemo:~$ python -m pip install arrow
```

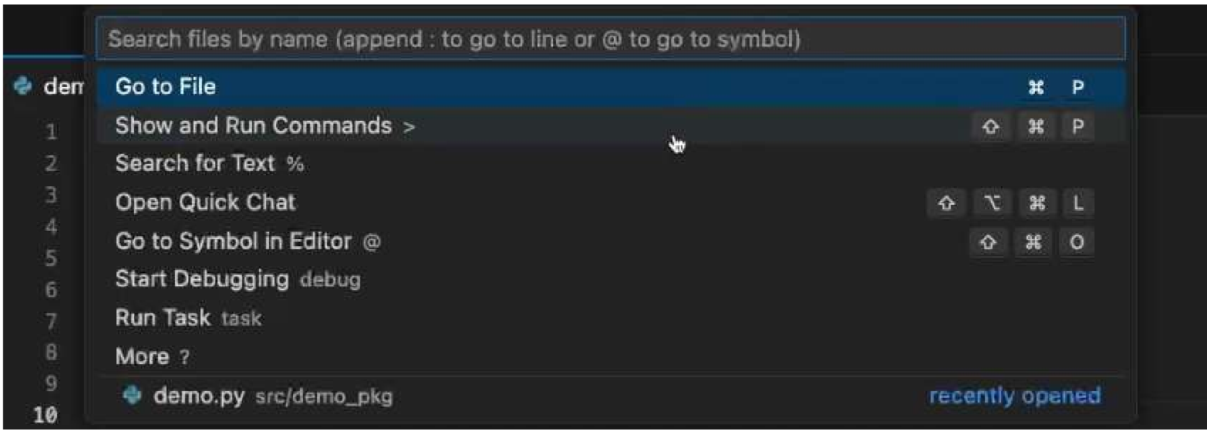
- This command invokes pip as a module to download and installed arrow
- Using python -m inside the command statements as it helps us install the package that maybe not compatibel with the pip version
-
```
(venv) rjekker@psdemo:~$ python -m pip uninstall arrow
```
- Used for uninstalling the package
- By using documentation from python.org you can view all the way in which python command is to be given

Source (Pluralsight) : https://app.pluralsight.com/ilx/video-courses/python-development-environments/course-overview

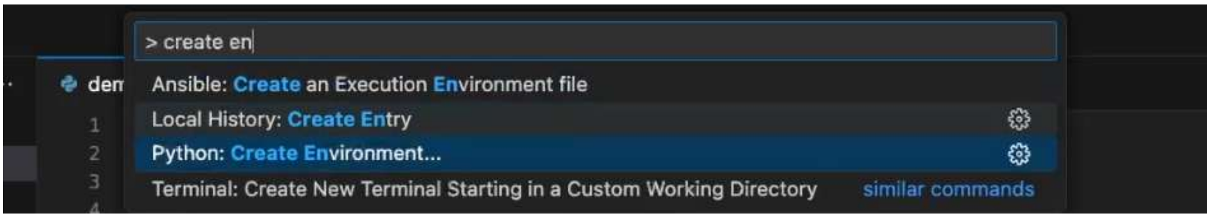
VS code

- VScode is one of the general editor (ide) and doesn't just focus on python , but also works with other languages like python , JavaScript etc
- To use and execute python project , start with installing Python extension Microsoft official.
- Then you can create an write python code on vscode.
- On vscode on the very bottom it shows the setup environment but initially it always shows some global environment
- We can setup our own virtual environment by following the given steps :

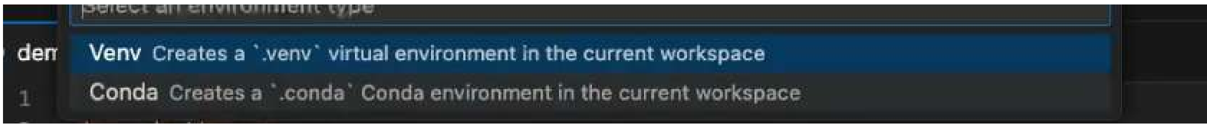
- On vscode click right on top



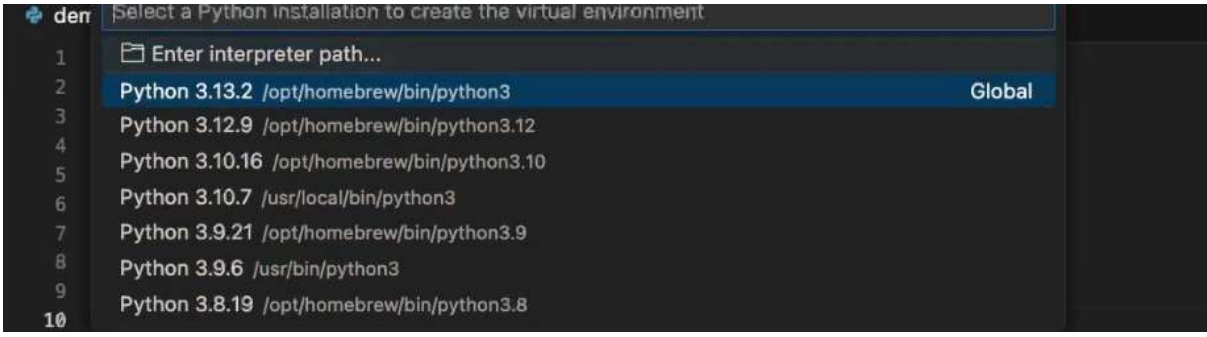
- Then selecting show and run command
- Then create your own environment by using Python create environment



- The specify to create venv



- Then select where to create the environment



Source (Pluralsight) : https://app.pluralsight.com/library/courses/amazon-bedrock-using-python

Jupyter or Colab free tier:

Jupyter

- To install and setup jupyter notebook on vscode we need to follow following steps :

- Step one : Open extensions and install jupyter notebook extension

- Step two : As jupyter notebook extension is being downloaded , now you can create a jupyter notebook file with (.ipynb) format

Source (Pluralsight) : https://app.pluralsight.com/library/courses/amazon-bedrock-using-python

OpenCV:

- OpenCV (Open Source Computer Vision Library) is an open-source library used for computer vision, image processing, and machine learning applications.
- It provides a comprehensive set of tools for analyzing, modifying, and understanding images and videos, making it widely used in both research and industry.
- Supports a wide range of computer vision tasks, including image processing, object detection, and feature extraction.
- Provides interfaces for multiple programming languages, such as Python, C++, and Java.

Numpy:

- Numpy is a python package in Python ,though it acts as a module when you import it inside your code
- It is a fundamental tool used for scientific computing , data analysis and machine learning and can also be used in computer vision for image representation
- Numpy provides a highly optimized N-dimensional array object (ndarray) which serves as a powerful replacement for standard Python lists
- To use numpy package within code , first step is import numpy inside code

```
import numpy as np
```

- Numpy is already installed as a part of python enviroment

- There are multiple functions that we can use within the code
- https://github.com/numpy/numpy/tree/main/numpy

How to use Numpy in computer vision:

- Image is a digital grid of numbers called pixels.
- Image digital grid is always being created after taking a note of each pixel
- Numpy itself cannot actually open a .jpg or .png filedirectly , instead we need to use it in conjunction with a python library called pillow or pil.
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Matplotlib

- Matplotlib is a Python library for creating static, interactive and animated visualizations from data.
- It provides flexible and customizable plotting functions that help in understanding data patterns, trends and relationships effectively.
- In computer vision , Matplotlib is a library that's actually going to allow us to plot and display image
- ```
  import matplotlib.pyplot as plt
  ```
- In order to display things inside the file or notebook , but in latest version we do not need it
  ```
  %matplotlib inline
  ```
-

Git:

- Git is a free, open-source distributed version control system designed to track changes in files and coordinate work among multiple people.
- It functions like a checkpoint save system for projects, allowing you to record history, revert to previous versions, and work together without losing data
- It consists of some commands also known as **Porcelain command**
  - git add
  - git push
  - git commit
  - git pull git push
  - git branch
  - git switch
  - git merge
  - git rebase
- It also consists of low level command also known as "**Plumbing command**"
- These are the commands on which porcelain commands are bulit
  - git cat-file
  ```

    - git hash-object
    - git count-objects
- We can image is a git as onion , so we have peel of the layers of onion git until we reach its core
- Common definition of git is Distributed Revision control system